Lecture 5: Portfolio Theory and Practice

Readings:

- BKM Ch 5 – 7, 25

Prof. Rujing Meng, HKU

1

What to do?

- What to do with your investment capital?
  - Hundreds of thousands of assets are traded on the market

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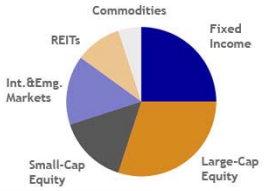
Construct a Portfolio



- Why?
  - Diversification

- How?
  - Markowitz modern portfolio theory

3

Why to Construct Portfolio
- Diversification Effect

The traditional theory on diversification "Don't put your eggs in one basket"
(Photo courtesy of www.moneywise.co.uk)

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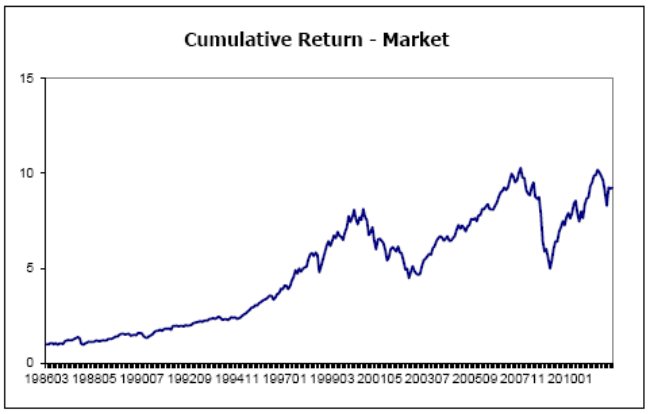
Diversification



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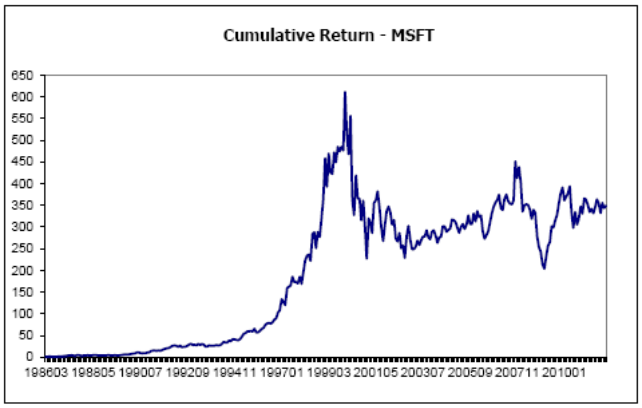
Diversification



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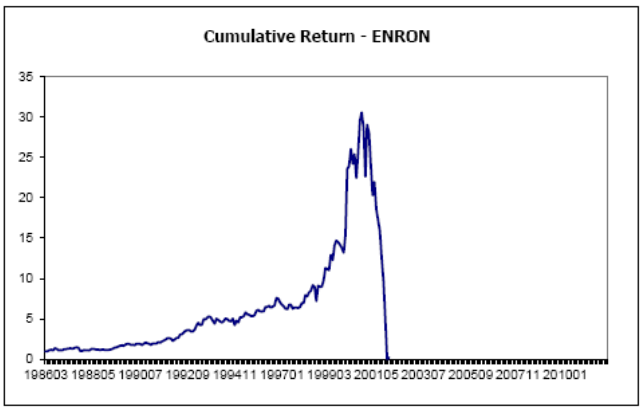
Diversification



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Diversification

- Sources of risks
  - Market risk (systematic risk, non-diversifiable risk)
    - Risks cannot be eliminated even by extensive diversification
    - Attributable to market-wide risk sources
  - Firm-specific risk (unique risk, nonsystematic risk, diversifiable risk)
    - Can be eliminated by diversification
- Diversification strategy
  - Naive diversification strategy
    - To hold equally weighted portfolios of several securities
  - Efficient diversification
    - To construct risk portfolios to provide the lowest possible risk for any given level of expected return.

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Naive diversification

- Naive diversification
  - To diversify into many more securities
  - When all risk is firm-specific, the portfolio risk can be reduced to virtually zero
  - When common sources affect all firms, even extensive diversification cannot eliminate risk

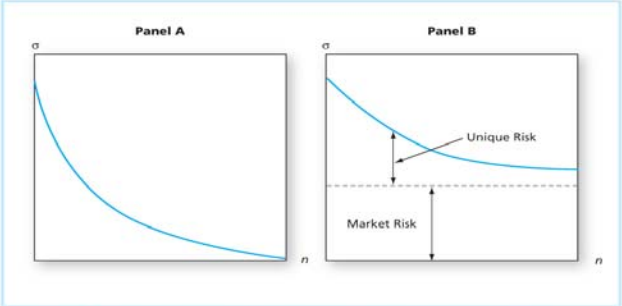

Figure 7.1 Portfolio risk as a function of the number of stocks in the portfolio
Panel A: All risk is firm specific. ***Panel B:*** Some risk is systematic, or marketwide.

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Naive diversification (cont.)

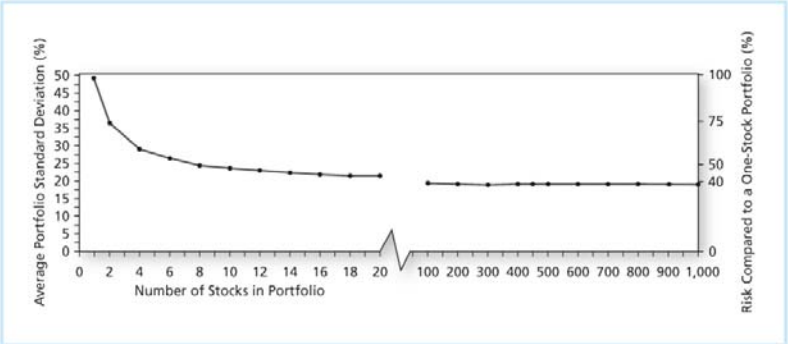

Figure 7.2 Portfolio diversification. The average standard deviation of returns of portfolios composed of only one stock was 49.2%. The average portfolio risk fell rapidly as the number of stocks included in the portfolio increased. In the limit, portfolio risk could be reduced to only 19.2%.

Source: From Meir Statman, "How Many Stocks Make a Diversified Portfolio?" *Journal of Financial and Quantitative Analysis* 22 (September 1987). Reprinted by permission.

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Naive diversification (cont.)

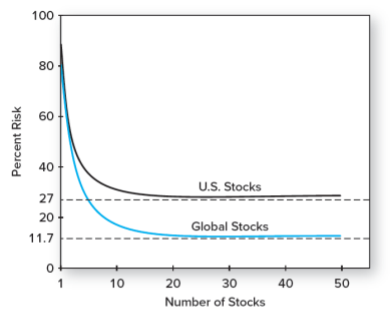

Figure 25.3 International diversification. Portfolio standard deviation as a percentage of the average standard deviation of a one-stock portfolio

Source: B. Solnik, "Why Not Diversify Internationally Rather Than Domestically?" *Financial Analysts Journal*, July/August 1974, pp. 48–54. Copyright 1995, CFA Institute. Reproduced and republished from *Financial Analysts Journal* with permission from the CFA Institute.

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Efficient diversification

- Portfolio expected return

Similarly, the expected return of a portfolio of assets can be calculated as the weighted average of the expected returns of each of the individual holdings, where the weights sum to one:

$$E[R_p] = \Sigma_i w_i E[R_i]$$

- Portfolio risk

The risk of a portfolio of assets (measured in terms of variance or $Var(R_p)$) is just the sum of every combination of assets' correlations multiplied by their respective standard deviations and their respective weights:

$$Var(R_p) = \Sigma_i \Sigma_j Corr(R_i,R_j) Stdev(R_i) Stdev(R_j) w_i w_j$$

and:

$$Stdev(R_p) = (Var(R_p))^{1/2}$$

Note that when i=j the correlation of an asset's return with itself is one and so the term becomes

$$Stdev(R_i) Stdev(R_i) w_i w_i = w_i^2 Var(R_i)$$

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Efficient diversification

- A portfolio of two risky assets
  - A mutual fund specializing in bond securities, denoted D
  - A stock fund, denoted E
- Risk and Return of the portfolio
  - $r_P = w_D r_D + w_E r_E$ (ex post)
  - $E(r_P) = w_D E(r_D) + w_E E(r_E)$ (ex ante)
  - $\sigma_P^2 = w_D^2\sigma_D^2 + w_E^2\sigma_E^2 + 2w_D w_E Cov(r_D,r_E)$
    $= w_D w_D Cov(r_D,r_D) + w_E w_E Cov(r_E,r_E) + 2w_D w_E Cov(r_D,r_E)$

where $Cov(r_D,r_E)$ = Covariance of returns for D and E

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Efficient diversification (cont.)

- Portfolio variance

A. Bordered covariance matrix (symmetric)

	w_D	w_E
w_D	$Cov(r_D,r_D)$	$Cov(r_D,r_E)$
w_E	$Cov(r_E,r_D)$	$Cov(r_E,r_E)$

B. Bordered-multiplied covariance matrix

	w_D	w_E
w_D	$w_D w_D Cov(r_D,r_D)$	$w_D w_E Cov(r_D,r_E)$
w_E	$w_E w_D Cov(r_E,r_D)$	$w_E w_E Cov(r_E,r_E)$

Portfolio variance

$w_D w_D Cov(r_D,r_D) + 2w_D w_E Cov(r_D,r_E) + w_E w_E Cov(r_E,r_E)$

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Concept Check

- Write out the covariance matrix of a portfolio of three-asset, X, Y, Z, with weights w_X, w_Y, and w_Z. Write out the portfolio variance
-

	w_X	w_Y	w_Z
w_X	$w_X w_X Cov(r_X,r_X)$	$w_X w_Y Cov(r_X,r_Y)$	$w_X w_Z Cov(r_X,r_Z)$
w_Y	$w_Y w_X Cov(r_Y,r_X)$	$w_Y w_Y Cov(r_Y,r_Y)$	$w_Y w_Z Cov(r_Y,r_Z)$
w_Z	$w_Z w_X Cov(r_Z,r_X)$	$w_Z w_Y Cov(r_Z,r_Y)$	$w_Z w_Z Cov(r_Z,r_Z)$

Portfolio variance:

$$w_X^2\sigma_X^2 + w_Y^2\sigma_Y^2 + w_Z^2\sigma_Z^2 + 2w_X w_Y Cov(r_X,r_Y) + 2w_X w_Z Cov(r_X,r_Z) + 2w_Y w_Z Cov(r_Y,r_Z)$$

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Efficient diversification (cont.)

- Portfolio risk depends on the correlation between the returns of the assets in the portfolio
- Covariance and the correlation coefficient provide a measure of the way returns of two assets vary
- $Cov(r_D,r_E) = \rho_{DE}\sigma_D\sigma_E$

  ρ_{DE} = Correlation coefficient of returns

  $-1 \leq \rho_{DE} \leq +1$

  If $\rho_{DE} = +1$, perfectly positively correlated

  If $\rho_{DE} = -1$, perfectly negatively correlated
- $\sigma_P^2 = w_D^2\sigma_D^2 + w_E^2\sigma_E^2 + 2w_D w_E Cov(r_D,r_E)$

  $= w_D^2\sigma_D^2 + w_E^2\sigma_E^2 + 2w_D w_E \rho_{DE}\sigma_D\sigma_E$

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Efficient diversification (cont.)

- Diversification effect
  - $\sigma_P^2 = w_D^2\sigma_D^2 + w_E^2\sigma_E^2 + 2w_D w_E \rho_{DE}\sigma_D\sigma_E$
  - If $\rho_{DE} = 1$, $\sigma_P = w_D\sigma_D + w_E\sigma_E$, there is no diversification
  - If $\rho_{DE} < 1$, $\sigma_P < w_D\sigma_D + w_E\sigma_E$, there is diversification
  - If $\rho_{DE} = -1$, $\sigma_P = \text{Abs}(w_D\sigma_D - w_E\sigma_E)$, perfect hedge is possible
    - $\sigma_P = 0$, when $w_E = \dfrac{\sigma_D}{\sigma_D + \sigma_E} = 1 - w_D$

Portfolio of less than perfectly correlated assets always offer better risk-return opportunities than the individual component securities on their own

To add in assets with low correlation with the existing position (negative correlation assets are even better)

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Efficient diversification (cont.)

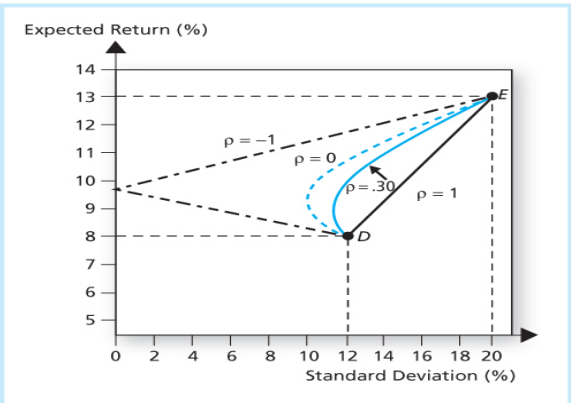

Figure 7.5 Portfolio expected return as a function of standard deviation

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How to Construct Portfolio
- Modern Portfolio Theory (MPT)

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Figure 5.4 The Normal Distribution

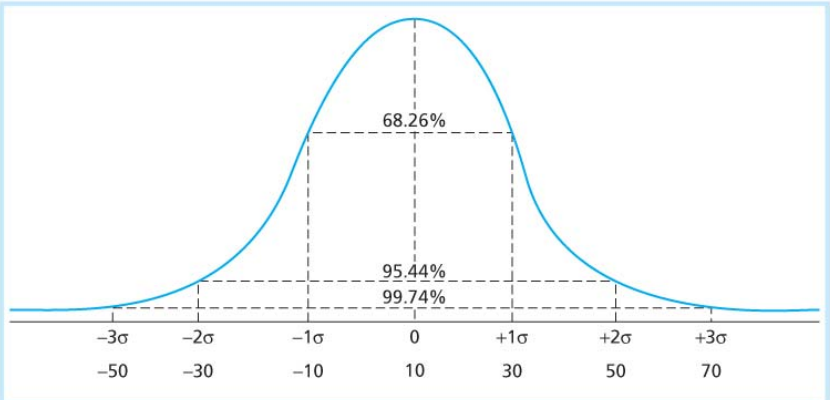

Figure 5.4 The normal distribution with mean 10% and standard deviation 20%.

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Figure 5.5A Normal and Skewed Distributions (mean = 6% SD = 17%)

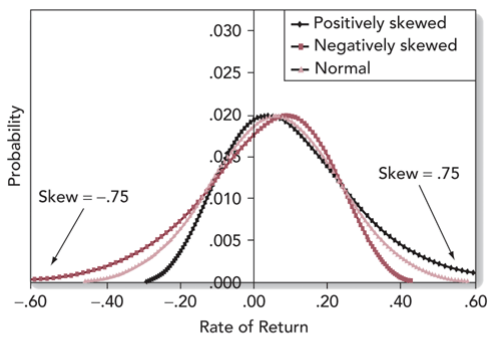

FIGURE 5.5A Normal and skewed distributions (mean = 6%, SD = 17%)

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Figure 5.5B Normal and Fat-Tailed Distributions (mean = .1, SD =.2)

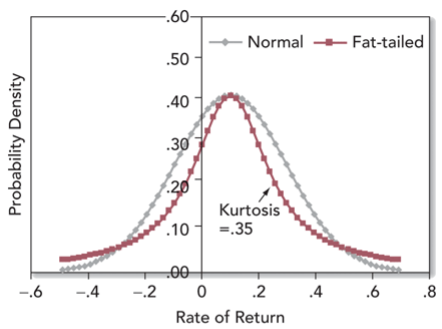

FIGURE 5.5B Normal and fat-tailed distributions (mean = .1, SD = .2)

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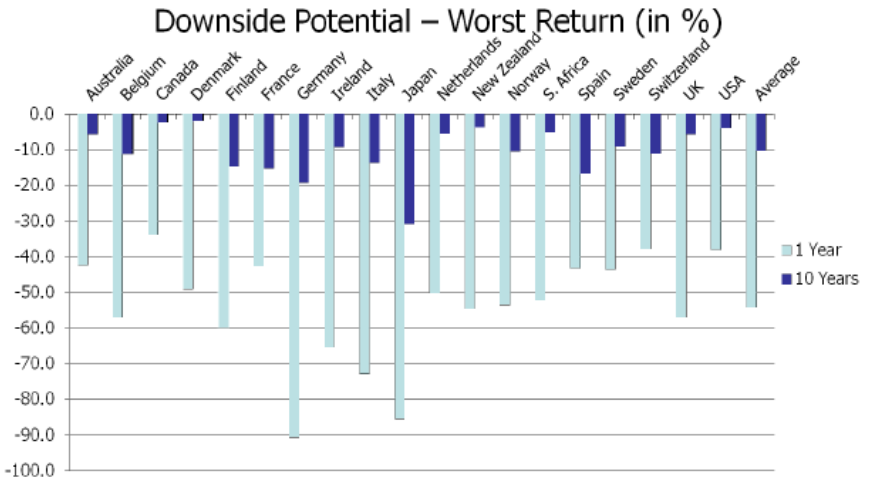

Sample: 1900 – 2009

Source: Estrada, "Stocks, bonds, Risk, and the Holding Period: An international Perspective," 2011.

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Capital Market Assumptions (Input Table)

- Data about the debt and equity funds

	Debt	Equity
Expected return E(r)	8%	13%
Std. Dev., σ	12%	20%
Correlation coefficient, ρ_{DE}	0.3	

- What's the return and risk of the portfolio?
  - $E(r_P) = w_D E(r_D) + w_E E(r_E) = 0.08w_D + 0.13w_E$
  - $\sigma_P^2 = w_D^2\sigma_D^2 + w_E^2\sigma_E^2 + 2w_D w_E \rho_{DE}\sigma_D\sigma_E$
    $= 0.0144w_D^2 + 0.04w_E^2 + 0.0144w_D w_E$
  - $w_D + w_E = 1$

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The Efficient Frontier With No Risk-Free Asset (Risky Investment Opportunity Set)

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Efficient frontier of risky assets

- Minimum-variance frontier of risky assets
  - A graph of the lowest possible variance that can be attained for a given portfolio expected return
- Efficient frontier of risky assets
  - The frontier that lies above the global minimum-variance portfolio
  - Portfolios on the efficient frontier are dominant: minimum variance given a specified expected return, and maximum return given a specified variance

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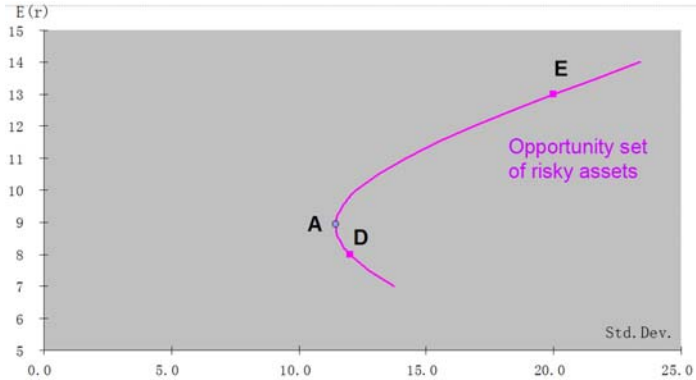

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Global Minimum Variance Portfolio

- To find out the weights to minimum portfolio variance
- Write out the variance as a function of w_D, and set its first derivative to zero
- $$w_D = \frac{\sigma_E^2 - \rho_{DE}\sigma_D\sigma_E}{\sigma_D^2 + \sigma_E^2 - 2\rho_{DE}\sigma_D\sigma_E} = 1 - w_E$$
- $w_D = 0.82$, $w_E = 0.18$,
- $E(r_P) = 8.90\%$, $\sigma_P = 11.45\%$

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The Risk-Free Asset and the Capital Allocation Line (CAL)

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Add in T-bills

- What if our risky assets are still confined to the bond and stock funds, but now we can also invest in risk-free T-bills yielding 5%?
- Capital allocation line
  - A straight line connecting a risk-free asset and a risky portfolio
  - To depict all the risk-return combinations available to investors
  - Slope: reward-to-variability ratio / Sharpe ratio
    - Incremental return per incremental risk
- Capital market line (passive strategy)
  - The CAL provided by 1-month T-bills and a broad index of common stocks

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Optimal CAL

- Which risk portfolio composed of the bond and stock funds is optimal to be combined with T-bills?
- Where is the optimal CAL?
- Is it the minimum-variance portfolio?

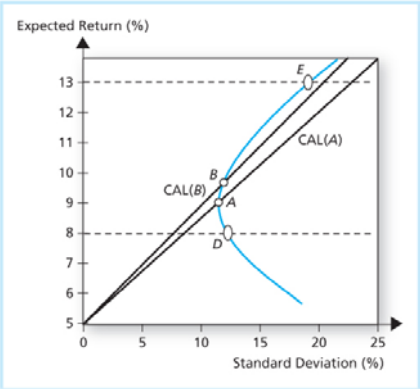

Figure 7.6 The opportunity set of the debt and equity funds and two feasible CALs

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Optimal CAL

- No! The CAL connecting risk-free rate and the minimum-variance portfolio is sub-optimal
- The optimal risk portfolio, *P*, should be the one maximizing the slope of the CAL. The slope is the Sharpe ratio.
- The tangency portfolio, P.
- The objective function is the Sharpe ratio : $S_P = \dfrac{E(r_P) - r_f}{\sigma_P}$

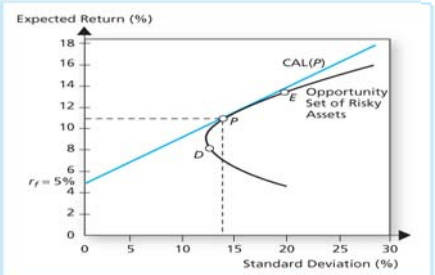

Figure 7.7 The opportunity set of the debt and equity funds with the optimal CAL and the optimal risky portfolio

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Optimal CAL

- $\underset{w_D}{Max}\, S_p = \dfrac{E(r_p) - r_f}{\sigma_p}$
  - $E(r_P) = 0.08w_D + 0.13w_E$
  - $\sigma_P = (0.0144w_D^2 + 0.04w_E^2 + 0.0144w_Dw_E)^{1/2}$
  - To represent S_P as a function of w_D, maximize S_P by selecting the w_D to make its first derivative equal to 0
  - $w_D^* = \dfrac{[E(r_D) - r_f]\sigma_E^2 - [E(r_E) - r_f]Cov(r_D, r_E)}{[E(r_D) - r_f]\sigma_E^2 + [E(r_E) - r_f]\sigma_D^2 - [E(r_D) - r_f + E(r_E) - r_f]Cov(r_D, r_E)}$
  - $w^*_D = 0.4$, $w^*_E = 0.6$
  - $E(r_P) = 11\%$, $\sigma_P = 14.2\%$
  - $S_P = 0.42$

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Optimal CAL

- Capital allocation
  - To invest a proportion of *y* in risky portfolio P, and *1-y* in risk-free asset F.
  - $E(r_P) = 11\%$, $\sigma_P = 14.2\%$
  - $r_f = 5\%$
  - Risk premium = 11% -5% = 6%
- *Complete* portfolio C
  - $E(r_C) = y*E(r_P) + (1-y)*r_f = r_f + y\,[E(r_P) - r_f] = 0.05 + y*0.06$
  - $\sigma_C = y*\sigma_P = 0.142y$
  - $y = \sigma_C/\sigma_P$

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Optimal CAL

- *Complete* portfolio C

$$E(r_C) = r_f + y[E(r_p) - r_f]$$

$$y = \frac{\sigma_C}{\sigma_P}$$

$$E(r_C) = r_f + \frac{\sigma_C}{\sigma_P}[E(r_P) - r_f]$$

$$= r_f + \left[\frac{E(r_p) - r_f}{\sigma_p}\right]\sigma_C$$

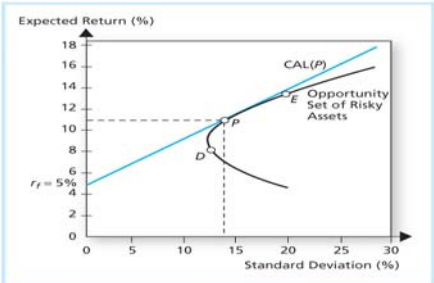

Figure 7.7 The opportunity set of the debt and equity funds with the optimal CAL and the optimal risky portfolio

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Optimal CAL

- The optimal CAL augments the risky investment opportunity set further towards northwest
- If investors agree on the risk-return properties of funds D and E, then they will face the common optimal CAL
- Each investor should choose his optimal complete portfolio on the common optimal CAL according to his/her risk tolerance

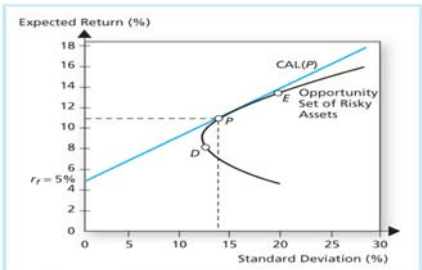

Figure 7.7 The opportunity set of the debt and equity funds with the optimal CAL and the optimal risky portfolio

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Optimal Complete Portfolio for Investors with Different Risk Tolerance

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Utility function

- To quantify investors' personal trade-offs between portfolio risk and expected return
- CFA scoring system
  - $U = E(r) - 0.5A\sigma^2$
  - U: utility, the score an investor assigns to his wealth
  - A: index of an investor's risk aversion
  - Utility is enhanced by high expected returns and diminished by high risk
  - The extent risk lowers utility depends on A
  - More risk-averse investors penalize risk more
  - Investors prefer higher utility level

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Each portfolio receives a utility score to assess the investor's risk/return trade off

Table 6.1

Available risky portfolios (Risk-free rate = 5%)

Portfolio	Risk Premium	Expected Return	Risk (SD)
L (low risk)	2%	7%	5%
M (medium risk)	4	9	10
H (high risk)	8	13	20

Investor Risk Aversion (*A*)	Utility Score of Portfolio *L* [$E(r) = .07; \sigma = .05$]	Utility Score of Portfolio *M* [$E(r) = .09; \sigma = .10$]	Utility Score of Portfolio *H* [$E(r) = .13; \sigma = .20$]
2.0	$.07 - ½ \times 2 \times .05^2 = .0675$	$.09 - ½ \times 2 \times .1^2 = .0800$	$\mathbf{.13 - ½ \times 2 \times .2^2 = .09}$
3.5	$.07 - ½ \times 3.5 \times .05^2 = .0656$	$\mathbf{.09 - ½ \times 3.5 \times .1^2 = .0725}$	$.13 - ½ \times 3.5 \times .2^2 = .06$
5.0	$.07 - ½ \times 5 \times .05^2 = .0638$	$\mathbf{.09 - ½ \times 5 \times .1^2 = .0650}$	$.13 - ½ \times 5 \times .2^2 = .03$

Table 6.2

Utility scores of alternative portfolios for investors with varying degrees of risk aversion

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Indifference curve

- Indifference curve
  - To a specified investor, a portfolio with high risk and high return may be equally attractive to a portfolio with lower risk and lower return
  - Indifference curve connects all portfolio points with the same utility value
- Example (A=4)

Expected Return, $E(r)$	Standard Deviation, σ	Utility = $E(r) - ½ A\sigma^2$
.10	.200	$.10 - .5 \times 4 \times .04 = .02$
.15	.255	$.15 - .5 \times 4 \times .065 = .02$
.20	.300	$.20 - .5 \times 4 \times .09 = .02$
.25	.339	$.25 - .5 \times 4 \times .115 = .02$

TABLE 6.3

Utility values of possible portfolios for investor with risk aversion, $A = 4$

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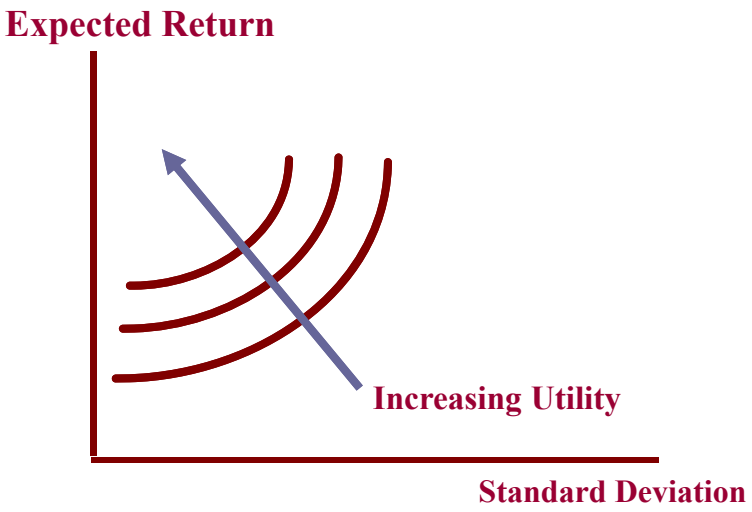
Indifference Curves



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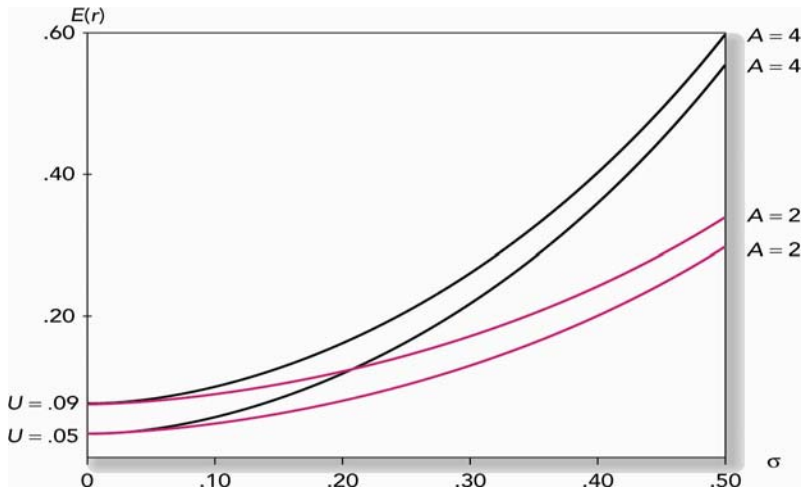
Indifference curve



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Optimal Complete Portfolio

- To find out the indifference curve which has the highest utility, which still has intersection with available investment set, CAL
- The tangent point of CAL with the indifference curve

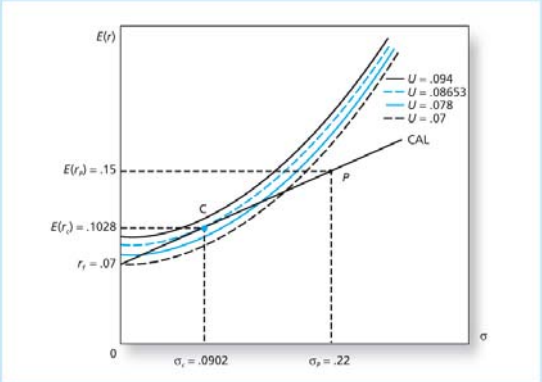

Figure 6.8 Finding the optimal complete portfolio by using indifference curves

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Optimal Complete Portfolio

- Utility maximization

$$\underset{y}{Max}\, U = E(r_C) - 0.5A\sigma_C^2 = [yE(r_P) + (1-y)r_f] - 0.5A[y\sigma_P]^2$$

$$F.O.C.:$$

$$\frac{\partial U}{\partial y} = [E(r_P) - r_f] - A\sigma_P^2 y = 0$$

$$y^* = \frac{E(r_P) - r_f}{A\sigma_P^2}$$

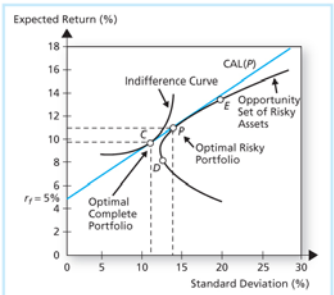

Figure 7.8 Determination of the optimal complete portfolio

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Optimal complete portfolio

- An investor with A=4

$$y^* = \frac{E(r_P) - r_f}{A\sigma_P^2} = \frac{0.11 - 0.05}{4 \times 0.142^2} = 74.39\%$$

$$E(r_C) = y^* E(r_P) + (1-y^*) r_f = 0.7439 \times 0.11 + 0.2561 \times 0.05 = 9.46\%$$

$$\sigma_C = y^* \sigma_P = 0.7439 \times 0.142 = 10.56\%$$

$$U = E(r_C) - 0.5A\sigma_C^2$$
$$= 0.0946 - 0.5 \times 4 \times 0.1056^2 = 0.0723$$

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Optimal complete portfolio

- An investor with A=4
  - To invest 74.39% in Portfolio P
    - % in bonds: 0.4*74.39% = 29.76%
    - % in stocks: 0.6*74.39% = 44.63%
  - To invest the remaining 25.61% in T-bills

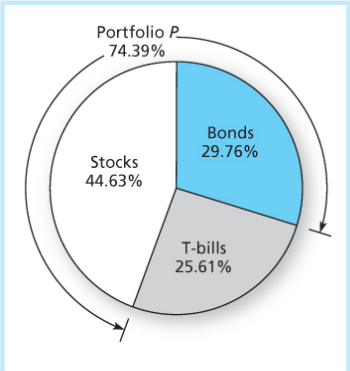

Figure 7.9 The proportions of the optimal complete portfolio

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Optimal complete portfolio

- Another more risk averse investor: A=8

$$y^* = \frac{E(r_P) - r_f}{A\sigma_P^2} = \frac{0.11 - 0.05}{8 \times 0.142^2} = 37.20\%$$

$$E(r_C) = y^* E(r_P) + (1 - y^*) r_f = 0.3720 \times 0.11 + 0.6280 \times 0.05 = 7.23\%$$

$$\sigma_C = y^* \sigma_P = 0.3720 \times 0.142 = 5.28\%$$

$$U = E(r_C) - 0.5A\sigma_C^2$$

$$= 0.0723 - 0.5 \times 8 \times 0.0528^2 = 0.0612$$

- A more **risk-averse** investor will allocate less to risky portfolio
- His indifference curve steeps upward faster than a risk-tolerant investor

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Steps to arrive at complete portfolio

1. Specify the return characteristics of all securities (expected returns, variances, covariances)
2. Establish the risky portfolio
   a. Calculate the optimal risky portfolio P
   b. Calculate the properties of Portfolio P
3. Allocate funds between risky portfolio P and the risk-free asset
   a. Calculate the fraction of the complete portfolio C allocated to Portfolio P
   b. Calculate the share of complete portfolio invested in each asset and in T-bills

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Concept check

- Two risky funds A & B, and T-bills

	Expected Return	Std. Dev.
A	10%	20%
B	30	60
T-bills	5	0

The correlation coefficient between funds A & B is -0.2

- Draw the opportunity set of funds A and B
- Calculate the global minimum-variance portfolio
- Find the optimal risky portfolio P, and its expected return & Std. Dev.
- Find the slope of the CAL supported by T-bills & P
- How much will an investor with A = 5 invest in funds A, B, and T-bills?

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Answers for the concept check

- Opportunity set
  - $E(r_P) = 0.10w_A + 0.30w_B$
  - $\sigma_P^2 = 0.04w_A^2 + 0.36w_B^2 - 0.048w_Aw_B$
  - $w_A + w_B = 1$
  - For each w_A, we can find corresponding $E(r_P)$ and σ_P so as to draw the opportunity set
- Global minimum-variance portfolio
  - To present σ_P^2 as a function of w_A
  - $\sigma_P^2 = 0.448w_A^2 - 0.768w_A + 0.36$
  - $w_A^* = 85.71\%$, $w_B^* = 14.29\%$, $E(r_P) = 12.86\%$, $\sigma_P = 17.57\%$

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Answers (cont.)

- Optimal risky portfolio P
  - Given $E(r_P)$ and σ_P^2, calculate the tangency point C
  - W^*_A = [(0.10-0.05)0.60² – (0.30-0.05)(-0.0240)] / [(0.10-0.05)0.60² + (0.30-0.05)0.20² – (0.10-0.05+0.30-0.05)(-0.0240)] = 68.18%
  - $E(r_P)$ = 16.36% , σ_P = 21.13%
  - S = (0.1636-0.05)/0.2113 = 0.5376
- Complete portfolio C
  - For an investor with A=5,
  - $y^* = \frac{E(r_P) - r_f}{A\sigma_P^2} = \frac{0.1636 - 0.05}{5 \times 0.2113^2} = 0.5089$
  - $E(r_C)$ = 10.80% , σ_C = 10.78%
  - Investment in T-bills: 1 – y* = 49.11%
  - Investment in A: $y^*w^*_A$ = 50.89%x68.18%=34.70%
  - Investment in B: $y^*w^*_B$ = 50.89%x31.82%=16.19%

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Many risky assets

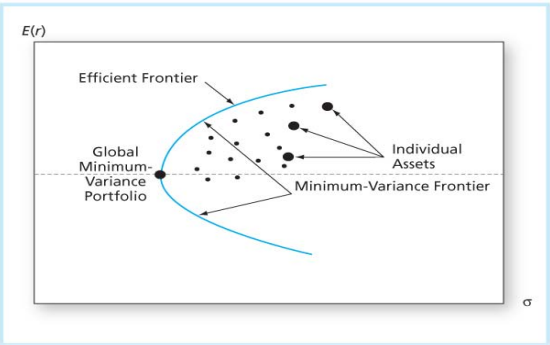

Figure 7.10 The minimum-variance frontier of risky assets

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Markowitz Portfolio Selection Model

- The first step is to determine the risk-return opportunities available.
- All portfolios that lie on the minimum-variance frontier from the global minimum-variance portfolio and upward provide the best risk-return combinations
- search for the CAL with the highest Sharpe ratio /reward-to-variability ratio

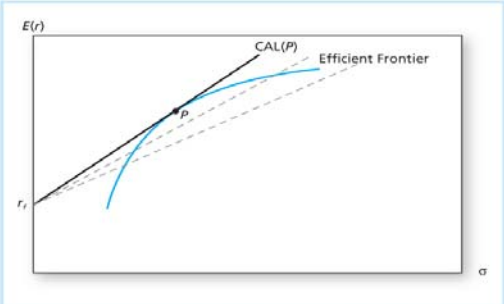

Figure 7.11 The efficient frontier of risky assets with the optimal CAL

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Markowitz Portfolio Selection Model

- Everyone invests in P, regardless of their degree of risk aversion.
  - More risk averse investors put more in the risk-free asset.
  - Less risk averse investors put more in P.

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Separation property

- The determination of the optimal risky portfolio, P, is purely technical
  - A portfolio manager will offer the same risky portfolio to all clients, regardless of their degree of risk aversion
  - CAL is uniform for all clients, to be the line starting from the risk-free asset to be tangent with the efficient frontier
- Determination the complete portfolio, C, to allocate between P and F, depends on personal preference

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Inputs to Markowitz Model

- In practice, different managers will offer different P
  - Different news
  - Different estimates on input list (expected returns, covariance matrix)
  - Different efficient frontiers
  - Different risky portfolio P
- The computerized optimization technique is the easiest part of the portfolio construction
- The real arena of competition is in sophisticated security analysis

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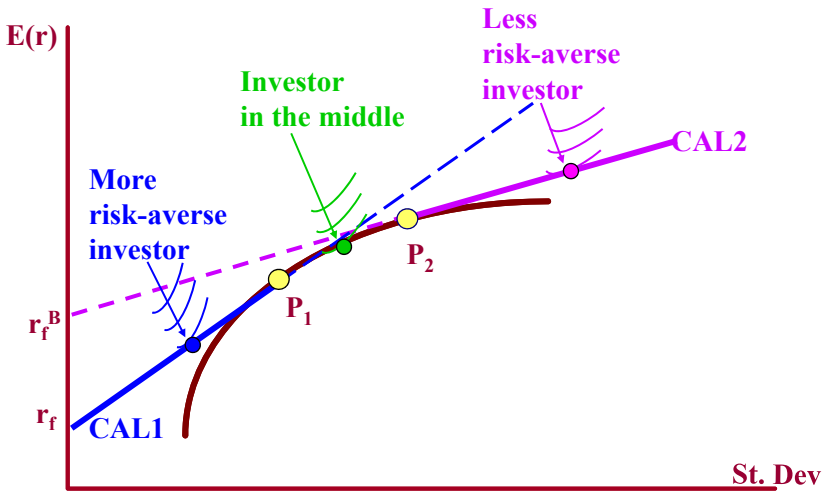
Leverage and Kinked CAL

- In practice, investors have to borrow at a premium to r_f



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Top-down analysis

- Capital allocation decision
  - Allocation between risk-free and risky assets
- Asset allocation decision
  - Choice of distribution across broad asset classes: stocks, bonds, real estate, foreign assets, etc
- Security selection decision
  - Which particular securities to hold within each asset class

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Top-down analysis (cont.)

- Most institutional investors follow a top-down approach
  - Capital and asset allocation decisions are made at a high organization level
  - Choice of specific securities is delegated to particular portfolio managers
- Individual investors follow a less-structured approach
  - Also give priority to broader allocation issues

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Figure 5.7 Nominal and Real Equity Returns Around the World, 1900-2000

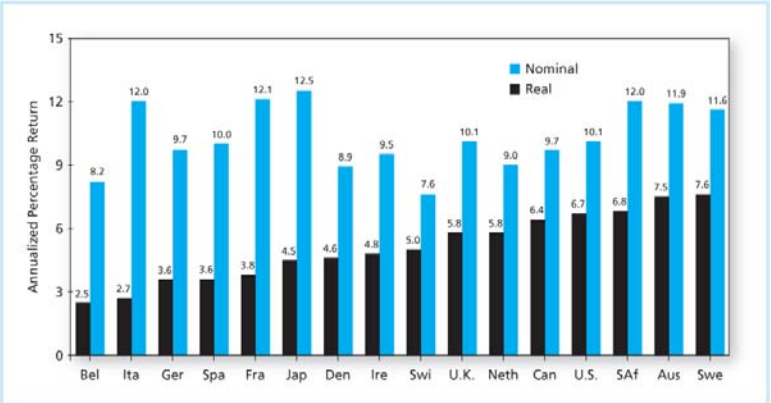

Figure 5.7 Nominal and real equity returns around the world, 1900–2000

Source: Elroy Dimson, Paul Marsh, and Mike Staunton, *Triumph of the Optimists: 101 Years of Global Investment Returns* (Princeton: Princeton University Press, 2002), p. 50. Reprinted by permission of the Princeton University Press.

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Figure 5.8 Standard Deviations of Real Equity and Bond Returns Around the World, 1900-2000

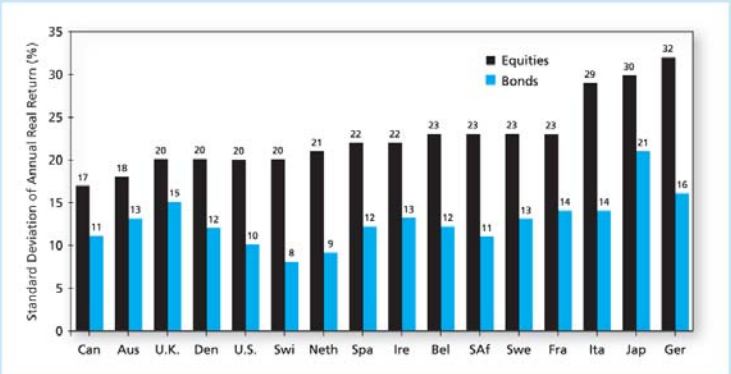

Figure 5.8 Standard deviations of real equity and bond returns around the world, 1900–2000

Source: Elroy Dimson, Paul Marsh, and Mike Staunton, *Triumph of the Optimists: 101 Years of Global Investment Returns* (Princeton: Princeton University Press, 2002), p. 61. Reprinted by permission of the Princeton University Press.

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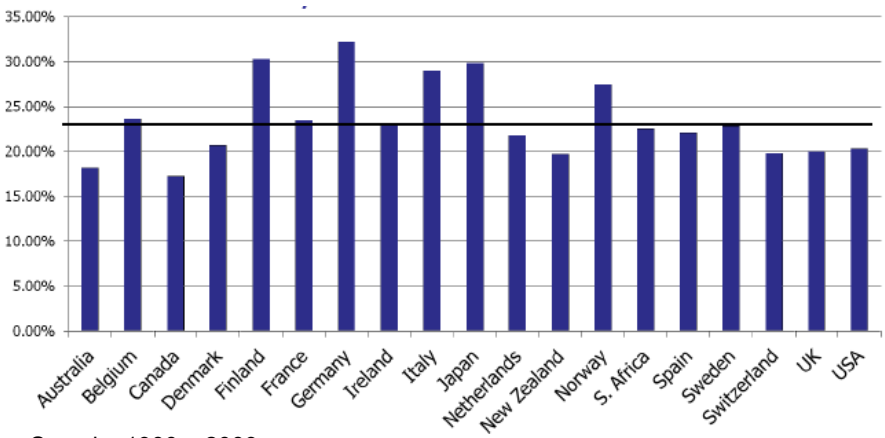
Annualized Volatility



Sample: 1900 – 2009

Source: Estrada, "Stocks, bonds, Risk, and the Holding Period: An international Perspective," 2011.

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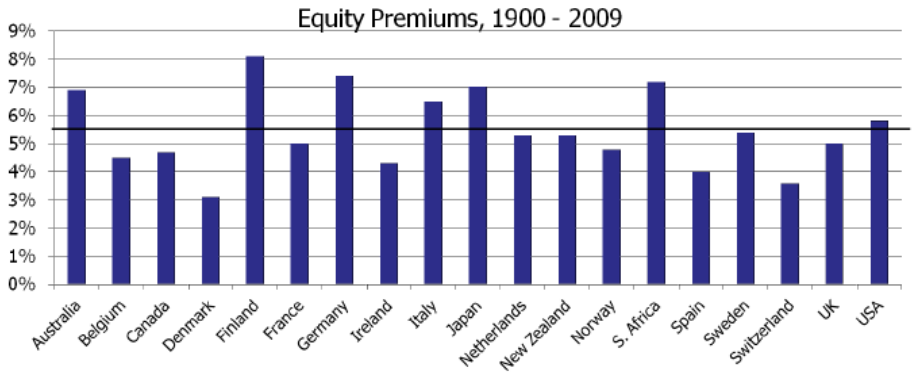
Consistently Positive Equity Premiums



Source: Estrada, "Stocks, bonds, Risk, and the Holding Period: An international Perspective," 2011.

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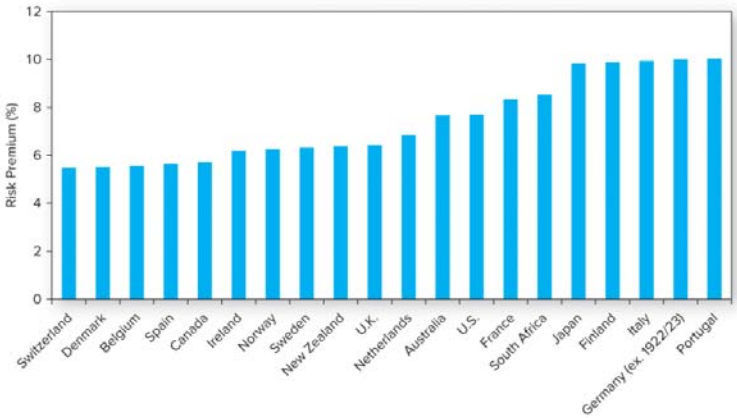

Figure 5.9 Average excess returns in 20 countries, 1900–2017

Note: The data for Germany omit the hyperinflation years 1922 and 1923.

Source: Authors' calculations using data from E. Dimson, P. R. Marsh, and M. Staunton, Triumph of the Optimists: 101 Years of Global Investment Returns (Princeton, NJ: Princeton University Press, 2002), with updates kindly provided by Triumph's authors.

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Top-down analysis (cont.)

- John Bogle, founder and late CEO of the Vanguard Group of Investment Companies
  - "The most fundamental decision of investing is the allocation of your assets: How much should you own in stock? How much should you own in bonds? How much should you own in cash reserves? …
  - "That decision [has been shown to account] for an astonishing 94% of the differences in total returns achieved by institutional managed pension funds…
  - "There is no reason to believe that the same relationship does not also hold true for individual investors."

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